A Controlled Benchmark of Visual State-Space Backbones with Domain-Shift and Boundary Analysis for Remote-Sensing Segmentation

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WHY A CONTROLLED BENCHMARK?

RESEARCH QUESTION

Do visual state-space backbones retain their advantages when only the encoder changes?

- Prior comparisons often change the decoder, feature interface, training schedule, and data pipeline together. Remote sensing segmentation needs both broad scene context and accurate roads, buildings, and object boundaries.
- Domain shift can expose failure modes that in-domain mIou alone does not explain.

CONTRIBUTIONS

- Strictly controlled comparison of VMamba, MambaVision, and Spatial-Mamba, plus CNN and Transformer reference encoders.
- Unified four-stage feature interface and the same lightweight U-Net decoder, loss, augmentations, schedule, and evaluation
- Joint analysis of in- domain accuracy, source-only domain shift, boundary versus interior error, and practical efficiency on two datasets.

STRICTLY CONTROLLED PROTOCOL

DATASETS AND SPLITS

Love DA

7 classes; official Urban 1,156/321 and Rural 1,366/410 train/val splits: All to All, U to R. and R to U

ISPRS Potsdam

6 classes: 38 tiles at 5 cm resolution; standard 24-train/14-val spit

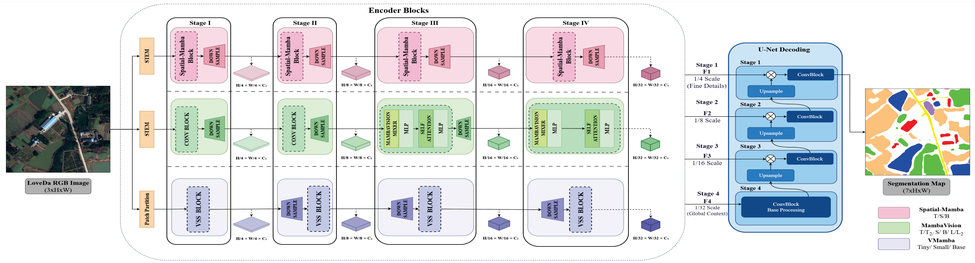
TRAINING

- 512x512 RGB patches: ImageNet normalization, batch size 4; 50k iterations
- AdamW; backbone LR 6e-5; decoder LR 3e-4; weight decay 0.05: Poly decity p 0.9.
- Loss Lovasz Focal 0.5x Boundary, boundary term uses 2-picel edge-dlated masks.
- Metric miou, OA, FI per-class Iou, boundary/interior mIou, Params, FLOPs, memory, and FPS.

CODE AND REPRODUCIBLE SETUP

github.com/Dineth14/ Mamba Segmentation
Fixed decoder, Fixed protocol, encoder only comparison

CONTROLLED BENCHMARK PIPELINE

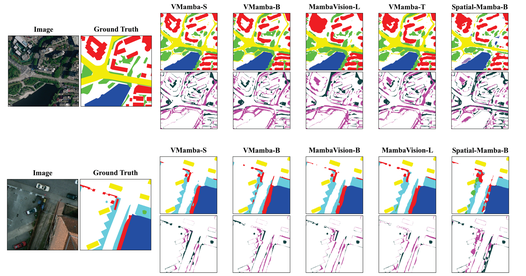


Every encoder is mapped to the same four-stage feature pyramid and the same lightweight U-Net decoder.

SELECTED QUANTITATIVE RESULTS

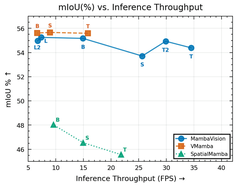
Type	Backbone	Params M	Flops G	Mem GB	FPS	LoveDA mIoU	LoveDA OA	LoveDA F1	U - R	R - U	Postdam mIoU	Postdam OA	Postdam F1
CNN	DeepLaby 3	36.0	105.3	0.40	18.52	43.01	59.14	64.20	30.36	39.98	75.09	89.06	84.44
Transformer	UNet Former	16.2	48.9	0.18	40.60	48.61	65.04	65.04	34.56	34.84	74.99	89.12	84.46
SSM	VMamba-Tiny	27.6	26.1	0.41	18.44	55.19	72.15	72.15	35.42	53.19	77.06	90.16	86.06
	VMamba-Small		42.5	0.49	10.50	55.66	73.49	70.69	40.62	53.52	77.59	90.43	86.46
SSM	VMamba-Base	81.7	67.3	0.70	7.77	55.61	72.89	70.54	35.82	55.50	77.43	90.61	86.19
SSM	Mambavision-T	36.8	44.9	0.19	40.83	54.39	71.00	69.88	36.12	53.89	76.59	90.01	85.65
SSM	Mamba Vision T2	40.1	50.7	0.21	34.87	54.93	72.24	69.87	38.17	51.02	76.65	90.00	85.72
SSM	MambaVision-S	55.6	65.6	0.27	30.19	53.71	71.19	69.07	39.47	53.46	76.77	90.13	85.77
SSM	MambaVision-B	104.2	123.8	0.45	17.72	55.17	72.42	70.43	40.24	51.96	77.08	90.27	86.03
SSM	Mamba Vision-L	236.6	273.6	0.97	8.64	55.25	72.79	70.39	38.53	54.01	77.07	90.30	85.98
SSM	MambaVision-L2	250.2	297.5	1.03	7.83	54.97	71.54	70.17	41.74	54.62	76.99	90.18	85.85
SSM	SpatialMamba-T	30.7	33.3	0.21	25.76	45.56	65.31	61.85	33.10	45.30	68.23	84.78	79.46
SSM	Spatial-Mamba-S	47.7	46.7	0.27	17.64	46.54	65.57	62.88	29.72	43.71	69.21	85.48	80.18
SSM	Spatial-Mamba-8	100.5	92.3	0.52	10.97	48.03	66.35	64.19	35.23	46.55	70.00	85.92	80.81

QUALITATIVE RESULTS: ALIGNED ERROR COMPARISON



ACCURACT-EFFICIENCY

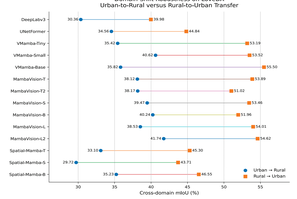
VMamba-Small:55.66 LoveDA | 77.59 Potsdam mIoU



40.83 FPS
MambaVision-T
LoveDA mIoU

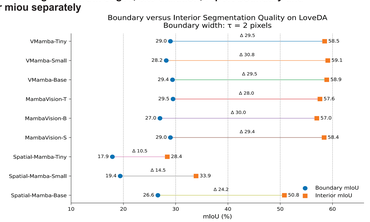
DOMAIN-SHIFT ANALYSIS

Source-only protocol train on one LoveDA domain, evaluate directly on the other-no adaptation



BOUNDARY ANALYSIS

Protocol Sobel ground truth edges, 3x3 dialation, report boundary and interior miou separately



TAKEAWAYS

- Visual SSM encoders offer favorable accuracy-efficiency trade-off's under this controlled setup
- Cross-domain performance is strongly asymmetric: R to U consistently exceeds U to R
- Future gains should prioritize boundary refinement, uncertainty modeling, and stronger multi-scale fusion over encoder scaling alone.